

The Gene Doctors

II PUC · Ch 10 · Biotechnology and its Applications

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



Why Change Plants?

"Plants whose genes have been changed are **GMOs**," said Prof. BAIO. "GM crops can tolerate drought and salt, need fewer pesticides, waste less after harvest and can be more nutritious — like **Golden rice**, rich in vitamin A."

💡 **REMEMBER**

Advantages of GM plants: stress tolerance, less pesticide use, fewer post-harvest losses, better nutrition.



The Bt Cotton Story

A bacterium, **Bacillus thuringiensis**, makes a **Cry protein** as an inactive **protoxin**. When a bollworm eats it, the alkaline gut activates the toxin, which makes holes in the gut lining. Genes like cryIAc and cryIIAb were put into cotton — **Bt cotton** protects itself!

💡 **REMEMBER**

cryIAc, cryIIAb → cotton bollworms · cryIAb → corn borer.



The Silent Treatment

A nematode, Meloidogyne incognita, attacked tobacco roots. Scientists used **RNA interference**: with Agrobacterium they put nematode genes into tobacco, which made sense and antisense RNA. This double-stranded RNA silenced the worm's mRNA, and the parasite could not survive.

💡 **REMEMBER**

RNAi = silencing of a specific mRNA by a complementary dsRNA.



Insulin from Bacteria

Insulin has two chains, A and B, joined by disulphide bridges. In 1983, the company **Eli Lilly** made chains A and B separately in E. coli and joined them to make **human insulin**. No more allergies from animal insulin!

💡 **REMEMBER**

Pro-insulin has an extra C-peptide that is removed in mature insulin.



Fixing a Gene

In 1990, a 4-year-old girl with **ADA deficiency** became the first **gene therapy** patient. Her lymphocytes were grown in the lab, given a working ADA gene using a retroviral vector, and returned to her body. It must be repeated, as the cells do not live for ever.

💡 **REMEMBER**

A permanent cure would introduce the gene into cells at an early embryonic stage.



Catching Disease Early

PCR copies tiny amounts of DNA, so it can find HIV even when the virus is still very low. **ELISA** works on antigen–antibody reactions. A radioactive **DNA probe** finds a mutated gene by autoradiography.

💡 **REMEMBER**

Early diagnosis means earlier and better treatment.



Special Animals

Transgenic animals carry a foreign gene; over 95% are mice. They help study diseases, test vaccine safety and make medicines like α-1-antitrypsin for emphysema. In 1997, **Rosie** the cow made milk with human α-lactalbumin — 2.4 grams per litre!

💡 **REMEMBER**

Transgenic animals: normal physiology, disease study, biological products, vaccine and chemical safety testing.



Rules and Rights

India's **GEAC** decides whether GM research and crops are safe. When a company tried to patent a basmati rice variety in 1997, people were angry: using a country's bio-resources without permission or payment is **biopiracy**. India amended its patent laws to stop it.

💡 **REMEMBER**

India has 27 documented varieties of basmati rice.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

GMOs p.1 · Golden rice p.1 · Bacillus thuringiensis p.2 · Cry protein p.2 · protoxin p.2 · Bt cotton p.2 · RNA interference p.3 · Eli Lilly p.4 · human insulin p.4 · ADA deficiency p.5 · gene therapy p.5 · PCR p.6 · ELISA p.6 · DNA probe p.6 · Transgenic animals p.7 · Rosie p.7 · GEAC p.8 · biopiracy p.8



QUICK QUIZ

Do you remember?

1. Bt toxin becomes active in the insect gut because of:

- Acidic pH
- Alkaline pH
- High temperature
- Enzymes of the plant

2. RNAi was used against:

- Bollworm
- Meloidogyne incognita
- Corn borer
- Aphids

3. Human insulin was first made in E. coli by:

- GEAC
- Eli Lilly
- IARI
- CDRI

4. The first gene therapy (1990) treated:

- Haemophilia
- ADA deficiency
- Diabetes
- Emphysema

5. Rosie the cow produced milk containing human:

- Insulin
- α-lactalbumin
- Interferon
- Antitrypsin

Answers: 1-Alkaline pH · 2-Meloidogyne incognita · 3-Eli Lilly · 4-ADA deficiency · 5-α-lactalbumin